Text as Data

5. Word Embeddings

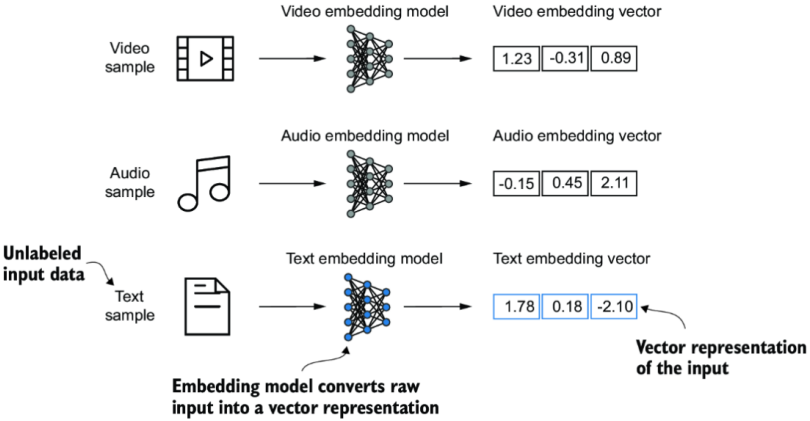

Figure 2.2 Deep learning models cannot process data formats like video, audio, and text in their raw form. Thus, we use an embedding model to transform this raw data into a dense vector representation that deep learning architectures can easily understand and process. Specifically, this figure illustrates the process of converting raw data into a three-dimensional numerical vector.

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  - the vector of features, $\boldsymbol{x}_i$, is itself a compressed representation of the unprocessed document $\mathcal{D}_i$.
- Further: the learned parameters $\hat{\theta}$ can also be understood as a **learned compressed representation of the whole dataset**:
  - it contains information about the training corpus, the text features, and the outcome classes.

Information in $\hat{\theta}$

Say we train a multinomial logistic regression on a bag-of-words representation $\boldsymbol{x}_i$ to predict classes $\boldsymbol{y}_i$:

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- θ is an interesting object. How can we use it?
- e.g.:
  - cluster the column vectors $\rightarrow$ which outcome classes are similar/related.
  - cluster the row vectors $\rightarrow$ which input features are similar/related.

$\hat{\theta}$ Contains Word Embeddings

θ = matrix of parameters learned from logit, relating words to outcomes.

- If $\boldsymbol{x}$ is a bag-of-words representation for a document consisting of a list of tokens $\{w_1, ..., w_t, ..., w_n\}$, we can write

$$\boldsymbol{x} = \frac{1}{n} \sum_{t=1}^{n} x_t$$

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- We can construct a **document vector**

$$\vec{\boldsymbol{d}} = \sum_{t=1}^{n_i} \theta_t$$

the sum of the n_y-dimensional word representations (the row vectors from above).

  - this is called a "continuous bag of words (CBOW)" representation (Goldberg 2017).
  - Note that $\vec{\boldsymbol{d}} = \theta \cdot \boldsymbol{x}$, we thus call θ a **word embedding matrix**.

Outline

Word Embedding without Neural Nets

Embedding Layers

Word Embedding with Neural Nets

Discussion: Word Embeddings

Word Embedding with Local Context

- Word2Vec and GloVe are two of the more well-known and commonly used models for producing word embeddings (vector representations of words).
  - the goal: represent the meaning of words by the neighboring words – their **contexts**.
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  - these models get good performance on a range of similarity, analogy, and prediction tasks.
- "You shall know a word by the company it keeps":
  - "He filled the **wampimuk**, passed it around and we all drunk some."
  - "We found a little, hairy **wampimuk** sleeping behind the tree."

Words and Contexts

Consider a **word-context matrix** $\boldsymbol{M}$:

- each row w represents a **word** (e.g. "income"), each column c represents a linguistic **context** in which a word can occur (e.g. "corporate ___ tax").
  - A matrix entry $\boldsymbol{M}_{[w,c]}$ quantifies the strength of association between a word and a context in a large corpus.

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  - A matrix entry $\boldsymbol{M}_{[w,c]}$ quantifies the strength of association between a word and a context in a large corpus.
- each word (row) $\boldsymbol{M}_{[w,:]}$ gives a distribution over contexts.
  - different definitions of contexts and different measures of association $\rightarrow$ different types of **word vectors**.
  - these vectors often have a **spatial interpretation** $\rightarrow$ geometric distances between word vectors reflect semantic distances between words.

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  - etc.
- Popular embeddings (word2vec and glove) generally use 5- or 10-word windows.
- Alternatives:
  - all words in the same sentence or same paragraph
  - syntactically connected words (from the parse tree)
  - Etc.

Defining an Association Measure

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- Downside: puts high weight on common contexts shared across many words (e.g., "the cat" will be weighted higher than "tabby cat")
- Better: **Point-wise mutual information** (PMI):

$$f_M(w, c) = \frac{\Pr(w, c)}{\Pr(w)\Pr(c)} = \frac{\frac{\#(w,c)}{n_D}}{\frac{\#(w)}{n_D}\frac{\#(c)}{n_D}} = \frac{n_D \#(w, c)}{\#(w)\#(c)}$$

where $\#(w)$ and $\#(c)$ are the corpus counts for w and c, respectively.

$\boldsymbol{M}$ is too high-dimensional

- $\boldsymbol{M}$ is $n_w \times n_c$
  - if c is drawn from from the vocabulary of a reasonably large corpus, the associated word vectors $\{v_1 = \boldsymbol{M}_{[w_1,:]}, v_2 = \boldsymbol{M}_{[w_2,:]}, \ldots\}$ are too high-dimensional to be useful.

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- Going back to dimension reduction: can use singular value decomposition (SVD):
  - factorize $\boldsymbol{M} \in \mathbb{R}^{n_w \times n_c}$ into a word matrix $\boldsymbol{W} \in \mathbb{R}^{n_w \times n_E}$ and context matrix $\boldsymbol{C} \in \mathbb{R}^{n_c \times n_E}$
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- $\boldsymbol{W}$ is the matrix of word vectors (word embeddings):
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  - dense, rather than sparse.
- **similarity measures between rows of $\boldsymbol{W}$ approximate similarity measures between rows of $\boldsymbol{M}$**

GloVe Embeddings

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Learn word vectors $\boldsymbol{w} = (w_1, ..., w_i, ..., w_{n_w})$, initialized randomly and $w_i \in (-1, 1)^{n_E}$, to solve

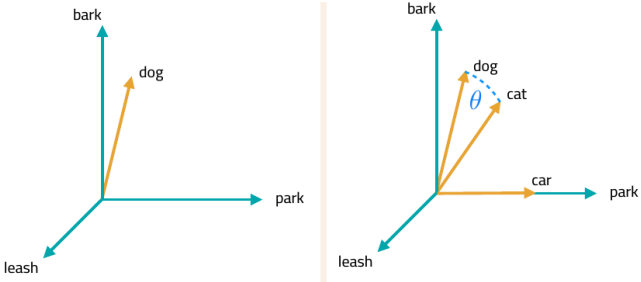
$$\min_{\boldsymbol{w}} \sum_{i,j} f(C_{ij}) \left(w_i^T w_j - \log(C_{ij}) \right)^2$$

where $f(\cdot)$ is a weighting function to down-weight frequent words.

- Minimizes **squared difference** between**:**
  - **dot product of word vectors,** $w_i^T w_j$
  - **empirical co-occurrence,** $\log(C_{ij})$
    [Arora et al (2016) put the PMI here instead of co-occurence counts]
- Intuitively: words that co-occur should have high correlation (dot product)

Word Similarity

- Once words are represented as vectors, we can use linear algebra to understand the relationships between words:
  - Words that are geometrically close to each other are similar: e.g. "dog" and "cat":



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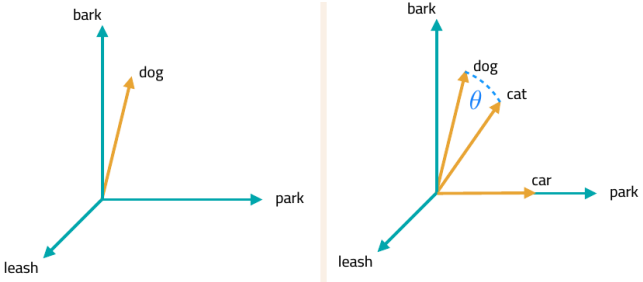

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- Thanks to linearity, can compute similarities between groups of words by averaging the groups.

Vector Geometry ↔ Semantic Meaning

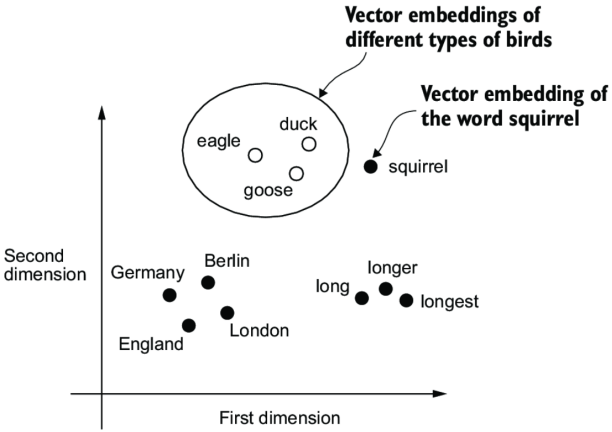
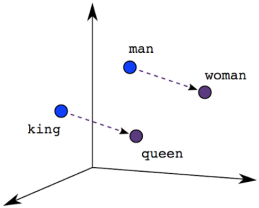


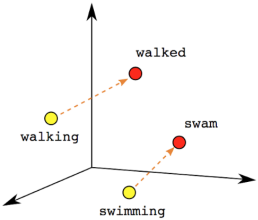
Figure 2.3 If word embeddings are two-dimensional, we can plot them in a two-dimensional scatterplot for visualization purposes as shown here. When using word embedding techniques, such as Word2Vec, words corresponding to similar concepts often appear close to each other in the embedding space. For instance, different types of birds appear closer to each other in the embedding space than in countries and cities.

Vector Directions ↔ Semantic Meaning

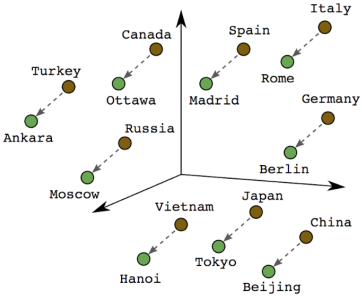
- Intriguingly, word embedding directions capture conceptual, analogical relationships between words:



Male-Female



Verb Tense



Country-Capital

Word Embeddings for Analogies

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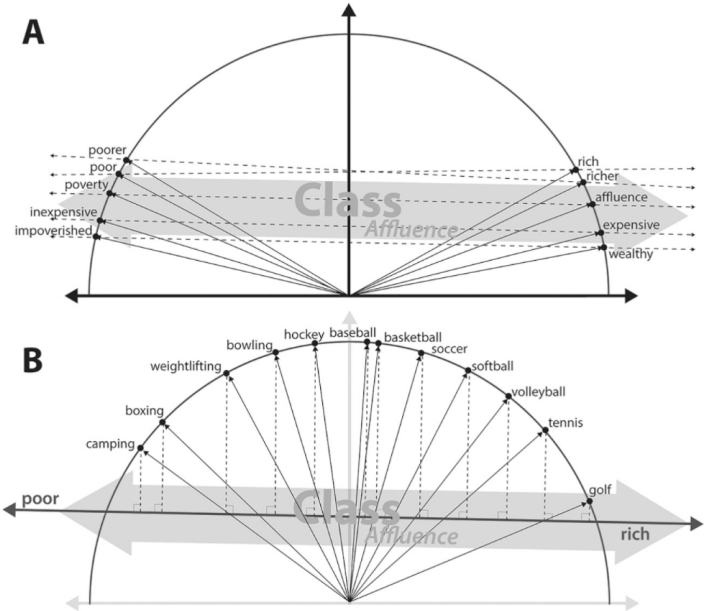
where V excludes (a_1, b_1, a_2).
- Often works better with normalized vectors (so that one long vector doesn't wash out the others)
- Levy and Goldberg (2014) recommend the following "CosMul" metric which tends to perform better:
$$\arg\max_{b_2 \in V} \frac{\cos(b_2, a_2)\cos(b_2, b_1)}{\cos(b_2, a_1) + \epsilon}$$
  - requires normalized, non-negative vectors (can transform using $(x+1)/2$)
  - ϵ is a small smoothing parameter.

Word Embeddings Capture Cultural Associations (Kozlowsky, Evans, and Taddy ASR 2019)



What is (and was) a person?
Evidence on historical mind perceptions from natural language

Elliott Ash, Dominik Stammbach, Kevin Tobia

***Cognition* (2023)**

Gray, Gray, & Wegner (*Science* 2007)

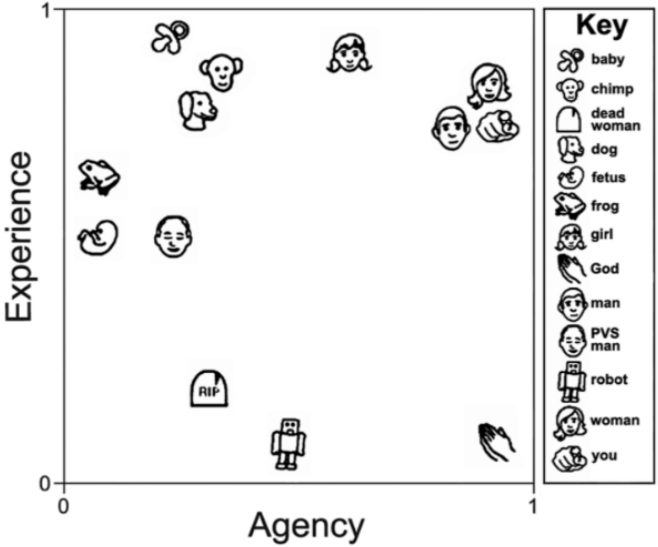


Fig. 1. Adjusted character factor scores on the dimensions of mind perception. PVS, persistent vegetative state.

Agency & Experience in Language: Measurement & Validation

Dimension	Word list
Agency	change cause aim intentional harm help decide control moral plan communicate think choose deliberate create guilty responsible act do express
Experience	pain pleasure enraged afraid hunger thirsty sad proud embarrassed joy angry happy conscious aware experience imagine awake suffer enjoy desire

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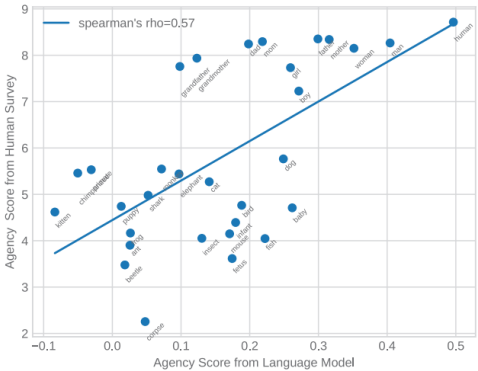
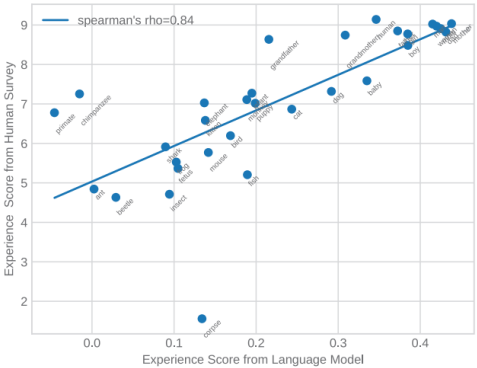
A. Agency



B. Experience



Gender Differences in Agency/Experience,1820s-2010s (COHA)

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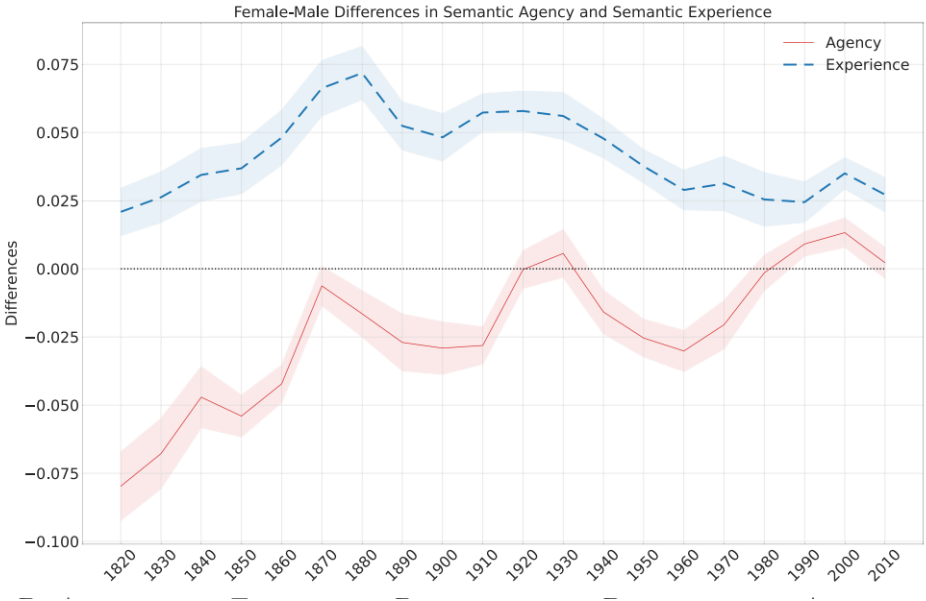

Domestic vs. Wild Animals,1820s-2010s (COHA)

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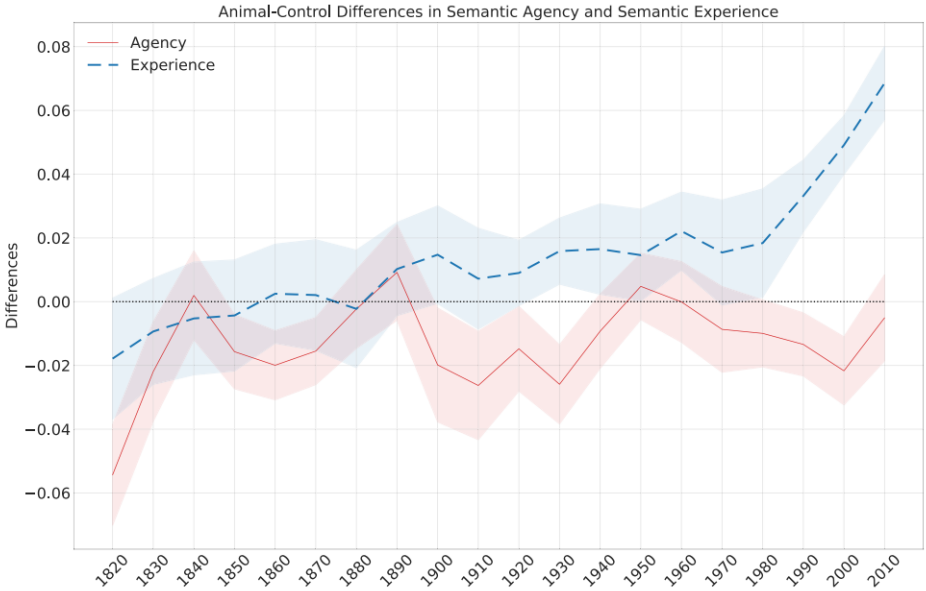


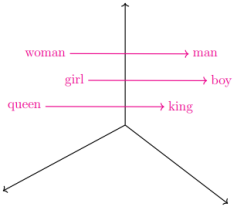
Gender Attitudes in the Judiciary:
Evidence from U.S. Circuit Courts

Elliott Ash, Daniel L. Chen, Arianna Ornaghi

***American Economic Journal: Applied Economics* (2023)**

Vector Directions $\leftrightarrow$ Semantic Dimensions

- The vector difference $\overrightarrow{man} - \overrightarrow{woman}$ identifies a step in "maleness" direction:



$$\overrightarrow{male} - \overrightarrow{female} = \frac{\sum_n \overrightarrow{male\ word_n}}{N_{male}} - \frac{\sum_n \overrightarrow{female\ word_n}}{N_{female}}$$

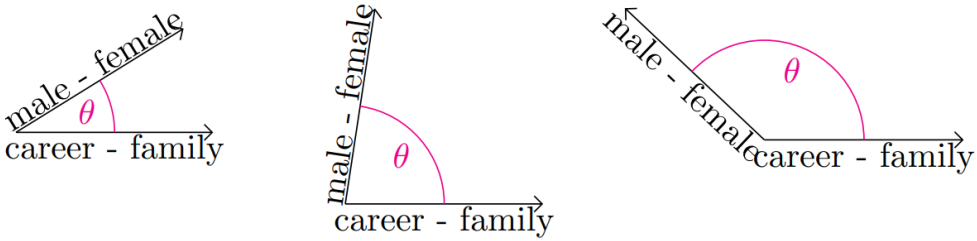
- Work-family dimension, defined by $\overrightarrow{work} - \overrightarrow{family}$

Application to U.S. Circuit Courts, 1887-2013

- learn separate embedding model for each judge, produce "gender stereotypes" measure for each based on stereotypical gender-career association.

Measuring Gender Stereotypes using Cosine of Angle between Dimensions

(a) Slant ≈ 1 **(b)** Slant ≈ 0 **(c)** Slant $\approx$ -1



Validation of Gender Stereotypes Measure

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5. Less gender stereoetypes $\leftrightarrow$ more use of gender-neutral pronouns (e.g. "he or she").
6. Less gender stereotypes $\leftrightarrow$ more empathic discussion of women's rights issues (human evaluation).

Does Judge Gender Stereotyping Matter?

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- shows that **gender attitudes measured in language really influence decisions**.

Outline

Word Embedding without Neural Nets

Embedding Layers

Word Embedding with Neural Nets

Discussion: Word Embeddings

Embeddings are critical to language models

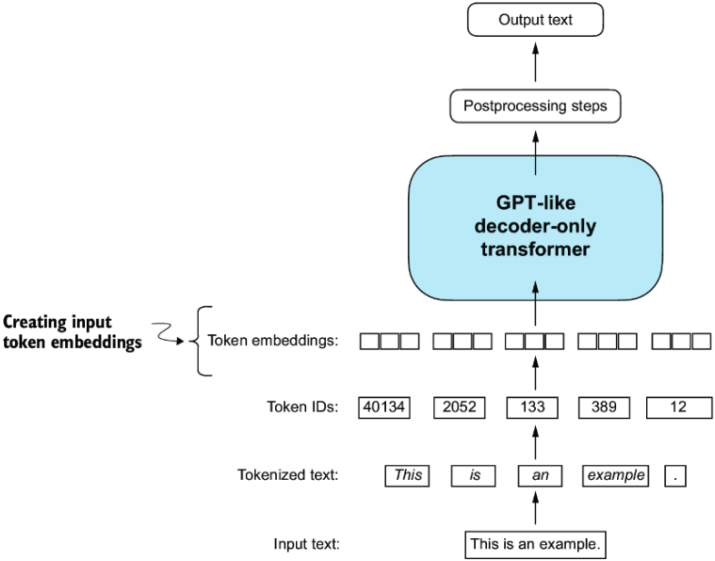

Figure 2.15 Preparation involves tokenizing text, converting text tokens to token IDs, and converting token IDs into embedding vectors. Here, we consider the previously created token IDs to create the token embedding vectors.

What is an Embedding?

- In a broad sense, **"embedding"** refers to a lower-dimensional dense vector representation of a higher-dimensional object.
  - in NLP, this higher-dimensional object will be a word, sentence, or document.

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  - sklearn CountVectorizer count vectors
- Embeddings:
  - PCA reductions of the word count vectors
  - LDA topic shares

Categorical Embeddings = dense representations of categorical variables

Say we have a binary classification problem with outcome Y:

- we have a high-dimensonal categorical variable (e.g. area of law with 1000 categories)
- including dummy variables A for each category in your ML model is computationally expensive.

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(2) is similar to what embedding layers do in neural nets.

In deep learning, an embedding layer is matrix multiplication:

$$\underbrace{h_1}_{n_E \times 1} = \underbrace{\omega_E}_{n_E \times n_w} \cdot \underbrace{x}_{n_x \times 1}$$

- x = a categorical variable (e.g., representing a word)
  - one-hot vector with a single item equaling one. Input to the embedding layer.
- ω_E = the matrix of learnable parameters.
- h_1 = the first hidden layer of the neural net
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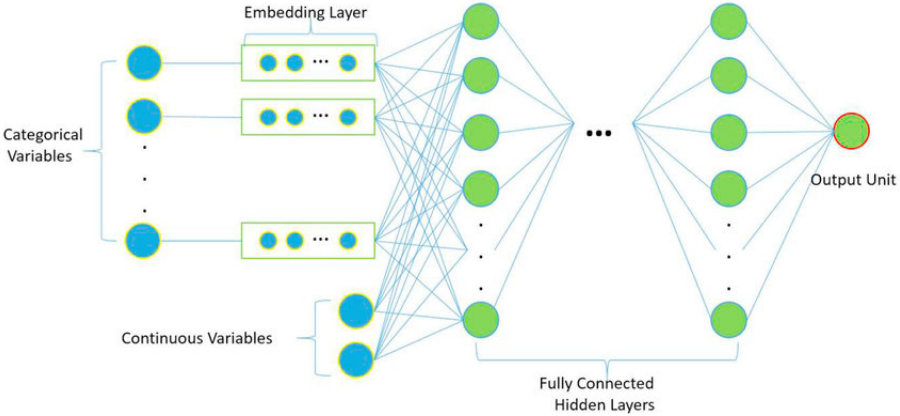

The embedding matrix ω_E encodes predictive information about the categories, has a spatial interpretation when projected to two dimensions.

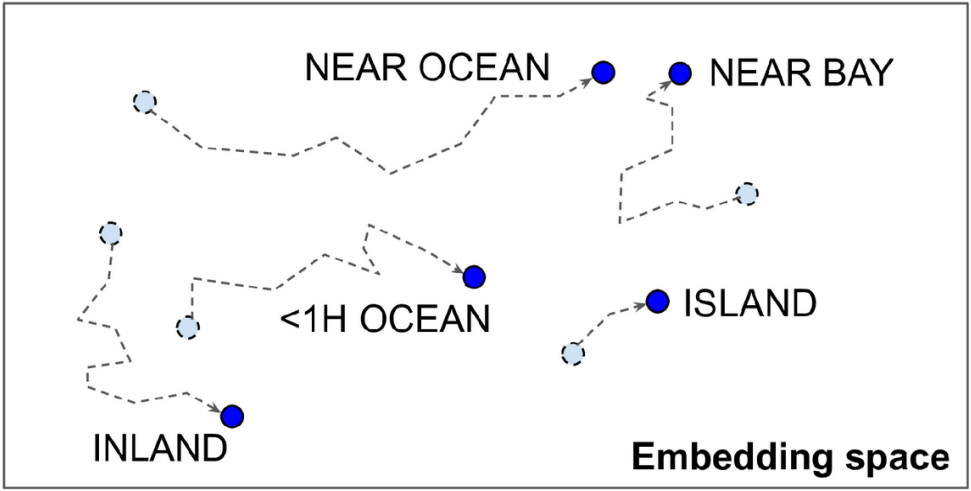

Figure 13-4. Embeddings will gradually improve during training

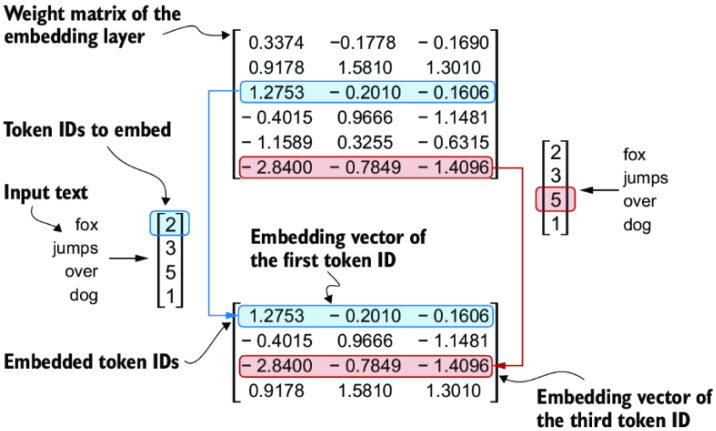

Figure 2.16 Embedding layers perform a lookup operation, retrieving the embedding vector corresponding to the token ID from the embedding layer's weight matrix. For instance, the embedding vector of the token ID 5 is the sixth row of the embedding layer weight matrix (it is the sixth instead of the fifth row because Python starts counting at 0). We assume that the token IDs were produced by the small vocabulary from section 2.3.

Embedding Layers versus Dense Layers

- An embedding layer is statistically equivalent to a fully-connected dense layer with one-hot vectors as input and linear activation.
  - dense layers might fit the data better, since you can use ReLU activation.
  - embedding layers are much faster; should use them when you have 10 or more categories.

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- Embedding layer replaces the list of sparse one-hot vectors with a list of n_E-dimensional ($n_E << n_w$) dense vectors

$$\boldsymbol{X} = \begin{bmatrix} x_1 & \dots & x_L \end{bmatrix}$$

where

$$\underbrace{x_j}_{n_E \times 1} = \underbrace{\boldsymbol{E}}_{n_E \times n_w} \cdot \underbrace{w_j}_{n_w \times 1}$$

  - $\boldsymbol{E}$ is a matrix of word vectors. The column associated with the word at j is selected by dot-product with one-hot vector w_j.

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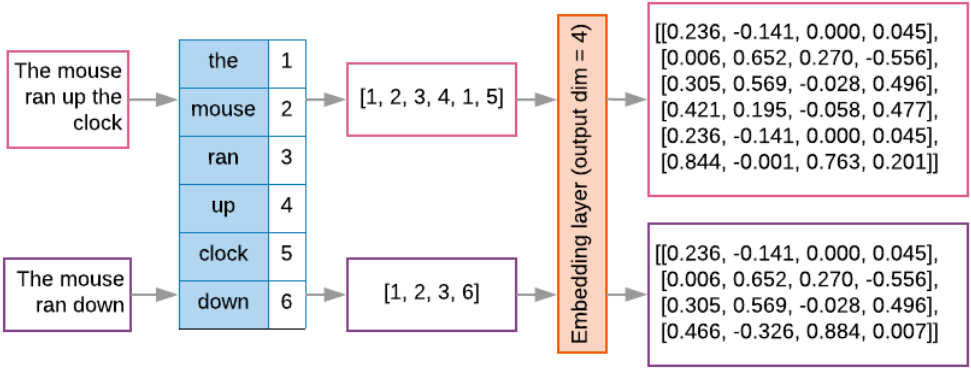
  - $\boldsymbol{E}$ is a matrix of word vectors. The column associated with the word at j is selected by dot-product with one-hot vector w_j.
- $\boldsymbol{X}$ is flattened into an $L * n_E$ vector for input to the next layer.

Illustration



Word2Vec

- "Word2Vec" is a neural net model that, instead of predicting some metadata (such as classifying topic labels), predicts the co-occurence of neighboring words.
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  - By comparing true instances of the word fox ("The quick brown **fox** jumps over the lazy dog")
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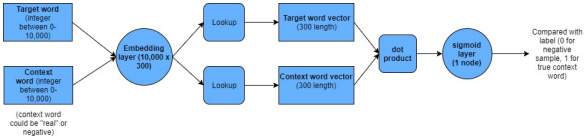
  - By comparing true instances of the word fox ("The quick brown **fox** jumps over the lazy dog")
  - to fake (randomly sampled) ones ("The prescription of **fox** is advised for this diagnosis")
- Word2Vec learns embedding vectors for the target word ("fox") and context words (neighbors of "fox") to distinguish true from false samples.

Word2Vec Negative Sampling Objective

The dataset is a collection of context pairs indexed by i:

- $y_i = 1$ means correct (it appeared in the corpus)
- $y_i = 0$ means incorrect (it was randomly drawn $\rightarrow$ **negative sample**).



- Both words are looked up in the same embedding matrix.
- The concatened embeddings $[\boldsymbol{w}; \boldsymbol{c}]$ are input to a dense layer (no activation) then to sigmoid output:

$$\hat{y}(w, c) = \text{sigmoid}(([\boldsymbol{w}; \boldsymbol{c}] \cdot \omega_0) \cdot \omega_1)$$

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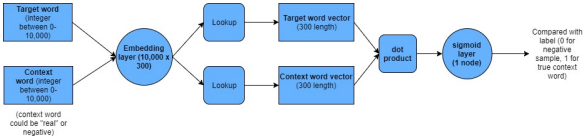

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- Word2Vec minimizes the binary cross-entropy

$$\min_{\boldsymbol{w}, \boldsymbol{c}, \omega} L(\boldsymbol{w}, \boldsymbol{c}, \omega) = -\sum_{i=1}^{n_D} [y_i \log \hat{y}_i(\boldsymbol{w}, \boldsymbol{c}, \omega) + [1 - y_i] \log(1 - \hat{y}_i(\boldsymbol{w}, \boldsymbol{c}, \omega))]$$

How does Word2Vec relate to the $\boldsymbol{M}$ matrix?

- Word2Vec produces embedding matrices $\boldsymbol{W}$ and $\boldsymbol{C}$.
  - generally, context embeddings are discarded after training.
- Levy and Goldberg (2014):
  - If we take $\tilde{\boldsymbol{M}} = \boldsymbol{W}\boldsymbol{C}'$, word2vec is equivalent to factorizing a matrix $\boldsymbol{M}$ with items

$$\boldsymbol{M}_{[w,c]} = \mathrm{PMI}(w, c) - \log a$$

where a is a constant calibrating the amount of negative sampling.

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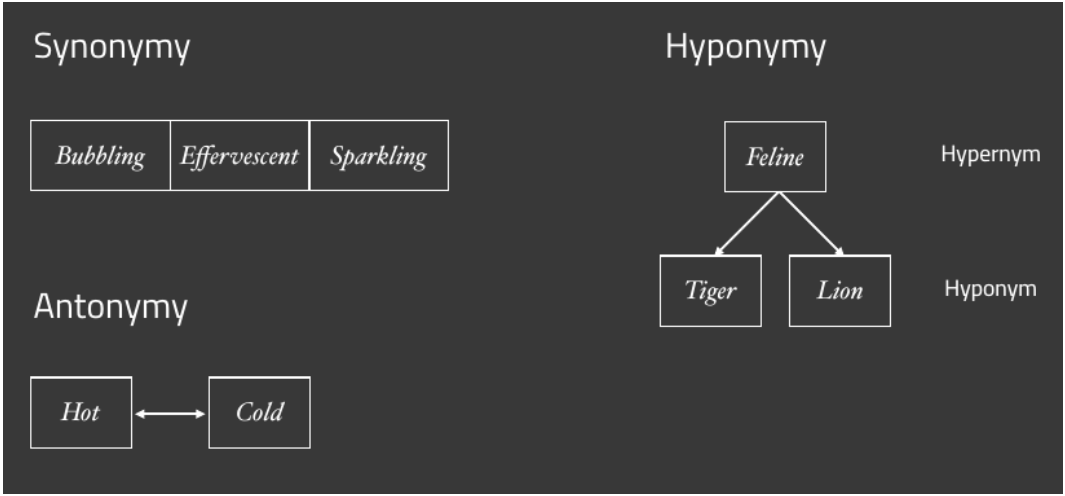

Similarity vs. Relatedness (Budansky and Hirst, 2006)

- Semantic **similarity**: words sharing salient attributes / features
  - synonymy (car / automobile)
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- Word embeddings will recover one or both of these relations, depending on how contexts and associated are constructed.

Most similar words to dog, depending on context window size

	2-word window	30-word window	
More paradigmatic	cat	kennel	More syntagmatic
	horse	puppy	
	fox	pet	
	pet	bitch	
	rabbit	terrier	
	pig	rottweiler	
	animal	canine	
	mongrel	cat	
	sheep	bark	
	pigeon	alsatian	

- Small windows pick up substitutable words; large windows pick up topics.

Pre-trained word embeddings

- In many settings (e.g. a small corpus), better to use pre-trained embeddings.
- e,g, spaCy's GloVe embeddings:
  - one million vocabulary entries
  - 300-dimensional vectors
  - trained on the Common Crawl corpus
- Can initialize models with pre-trained embeddings, can fine-tune as needed.

Parts of Speech and Phrases

- In the default model multiple senses of a word are merged.
  - e.g. “I like a bird” (verb) and “I am like a bird” (preposition).

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- The default model only works by word, but "new york $\neq$ "new" + "york"
  - can tokenize phrases together (see Week 2 lecture) before training.

The black sheep problem

- The trivial or obvious features of a word are not mentioned in standard corpora.
- For example, although most sheep are white, you rarely see the phrase "white sheep".
  - so word2vec sometimes tells you sim(black,sheep) $>$ sim(white,sheep).

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- For example, although most sheep are white, you rarely see the phrase "white sheep".
  - so word2vec sometimes tells you sim(black,sheep) $>$ sim(white,sheep).
- This is really important when interpreting results using embeddings to anayze beliefs/attitudes.
- Relatedly, antonyms are often rated similarly, have to be careful with that.

Review: NLP "Bias" is statistical bias

- Self-supervised learning algorithms like Word2Vec learn ***all*** dimensions of word associations; not just ones we are most interested in.
  - e.g., true expressions of attitudes or perceptions.

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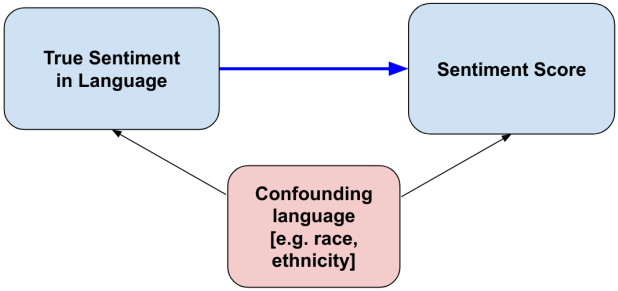
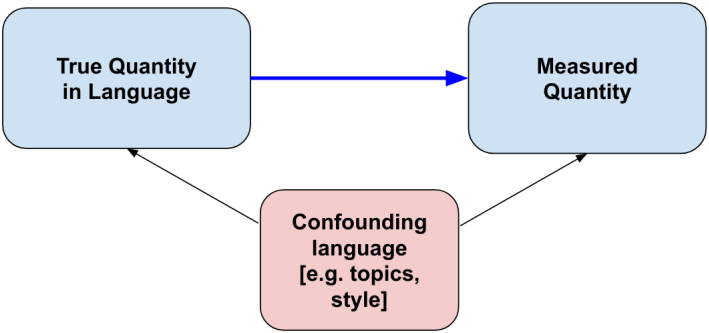
- Self-supervised learning algorithms like Word2Vec learn ***all*** dimensions of word associations; not just ones we are most interested in.
  - e.g., true expressions of attitudes or perceptions.
- e.g., using embeddings to scale social group words in a positive-to-negative dimension can learn correlated associations, not just sincere expressions of such attitudes:

```
text_to_sentiment("Let's go get Italian food")
2.0429166109
text_to_sentiment("Let's go get Chinese food")
1.4094033658
text_to_sentiment("Let's go get Mexican food")
0.3880198556
```

```
text_to_sentiment("My name is Emily")
2.2286179365
text_to_sentiment("My name is Heather")
1.3976291151
text_to_sentiment("My name is Yvette")
0.9846380213
text_to_sentiment("My name is Shaniqua")
-0.4704813178
```

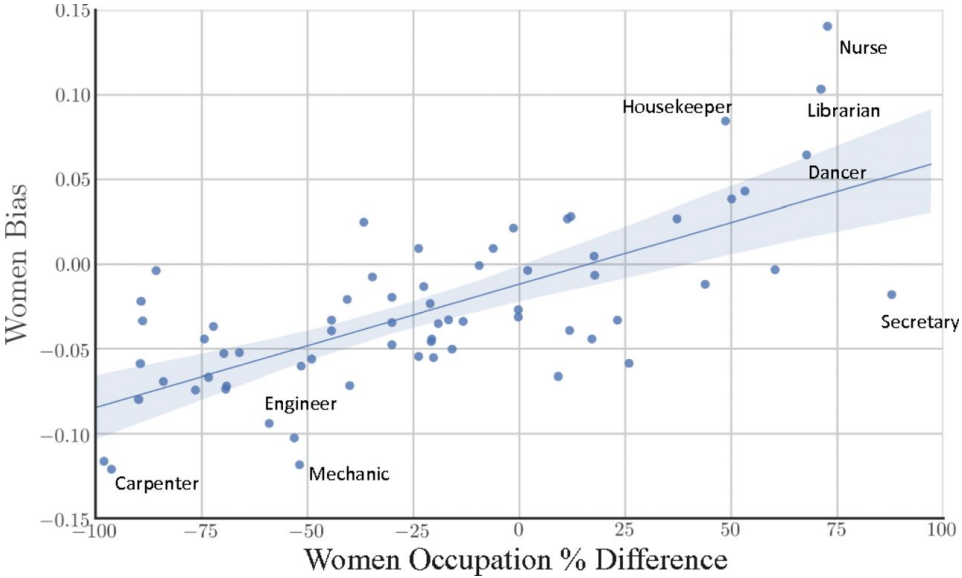


Self-Supervised Models Learn Confounders



- self-supervised language models like Word2Vec learn all lingusitic associations in language.
  - the measured associations might reflect attitudes/perceptions, or might reflect something else.

Garg, Schiebinger, Jurafsky, and Zou (PNAS 2018)



Women's occupation relative percentage vs. embedding bias in Google News vectors.

Steps for de-biasing

- Language features that are often confounded with the quantity of interest:
  - stopwords
  - named entities: person/organization/place names
- These can be dropped during pre-processing to reduce the influence of confounders in subsequent measurements.
- Can control for topic or style features or other potential confounders in regressions, or shuffle named entities.

De-Biasing Word Embeddings

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- Bolukbasi et al (NIPS 2016):
  - "Geometrically, **gender bias is first shown to be captured by a direction in the word embedding**."
  - "Second, gender neutral words are shown to be linearly separable from gender definition words in the word embedding."
  - "Using these properties, we provide a methodology for modifying an embedding to remove gender stereotypes, such as the association between the words receptionist and female, while maintaining desired associations such as between the words queen and female."

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- But: Gonen and Goldberg (2019):
  - ***"... we argue that this removal is superficial. While the bias is indeed substantially reduced according to the provided bias definition, the actual effect is mostly hiding the bias, not removing it. The gender bias information is still reflected in the distances between 'gender-neutralized' words in the debiased embeddings, and can be recovered from them..."***

Tokenizing for Word Embeddings

- capitalization?
- punctuation?
- stopwords/function-words?
- can add special tokens for start of sentence and end of sentence
- for out-of-vocab words, substitute a special token or replace with part-of-speech tag
  - or use FastText embeddings (more below)

"Enriching word vectors with subword information" (Bojanowski et al 2017)

- each word is represented as a bag of (hashed) character n-grams. (e.g., spicy = (spi, pic, icy)).
- learn embeddings for the character segments, and construct word embedding by summing over the segment embeddings

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- learn embeddings for the character segments, and construct word embedding by summing over the segment embeddings
- competitive with word2vec in standard tasks; better in some languages.
- produces good embeddings for unseen words.

Standard word embeddings (e.g. word2vec/glove) have a number of limitations:

- **polysemy**: you get one vector for multiple senses of a word (e.g. "**glass** of water" vs "window **glass**")

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- **polysemy**: you get one vector for multiple senses of a word (e.g. "**glass** of water" vs "window **glass**")
- **rare words**: a word that shows up just once or twice won't be well-defined
- **n-grams**: does not produce embeddings for multi-word phrases

Scientists attending ACL work on **cutting edge** research in NLP

Petrichor: the earthy scent produce when rain falls on dry soil

Roger Federer won the first **set**^{NN} of the match

Standard word embeddings (e.g. word2vec/glove) have a number of limitations:

- **polysemy**: you get one vector for multiple senses of a word (e.g. "**glass** of water" vs "window **glass**")
- **rare words**: a word that shows up just once or twice won't be well-defined
- **n-grams**: does not produce embeddings for multi-word phrases

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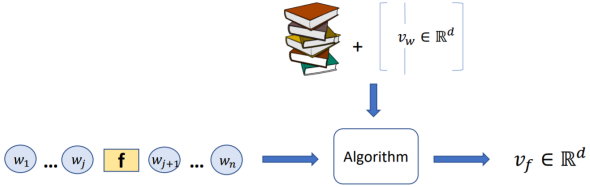
- Goal of Khodak et al (2018): produce embeddings "a la carte" given a context:

Given: Text corpus and high quality word embeddings trained on it



Input: A feature in context(s)

Output: Good quality embedding for the feature

A la carte embeddings

- Given a target word f and its context c, define

$$v_f^{avg} = \frac{1}{|c|} \sum_{w \in c} v_w$$

  the average vector for the words in the context.
- Arora et al (2018) prove that for vectors produced by a generative language model, there exists a matrix A such that

$$v_f \approx A v_f^{avg}$$

A la carte embeddings

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- The "induction matrix" A can be learned with a least-squares (linear regression) objective

$$A^* = \arg\min_A \sum_w |v_w - A v_w^{avg}|_2^2$$

  where w indexes over all the tokens in the corpus.
- empirically:

$$\text{cosine}(v_f, A^* v_f^{avg}) \geq 0.9$$